

Computational Wind Engineering

Turbulence modelling

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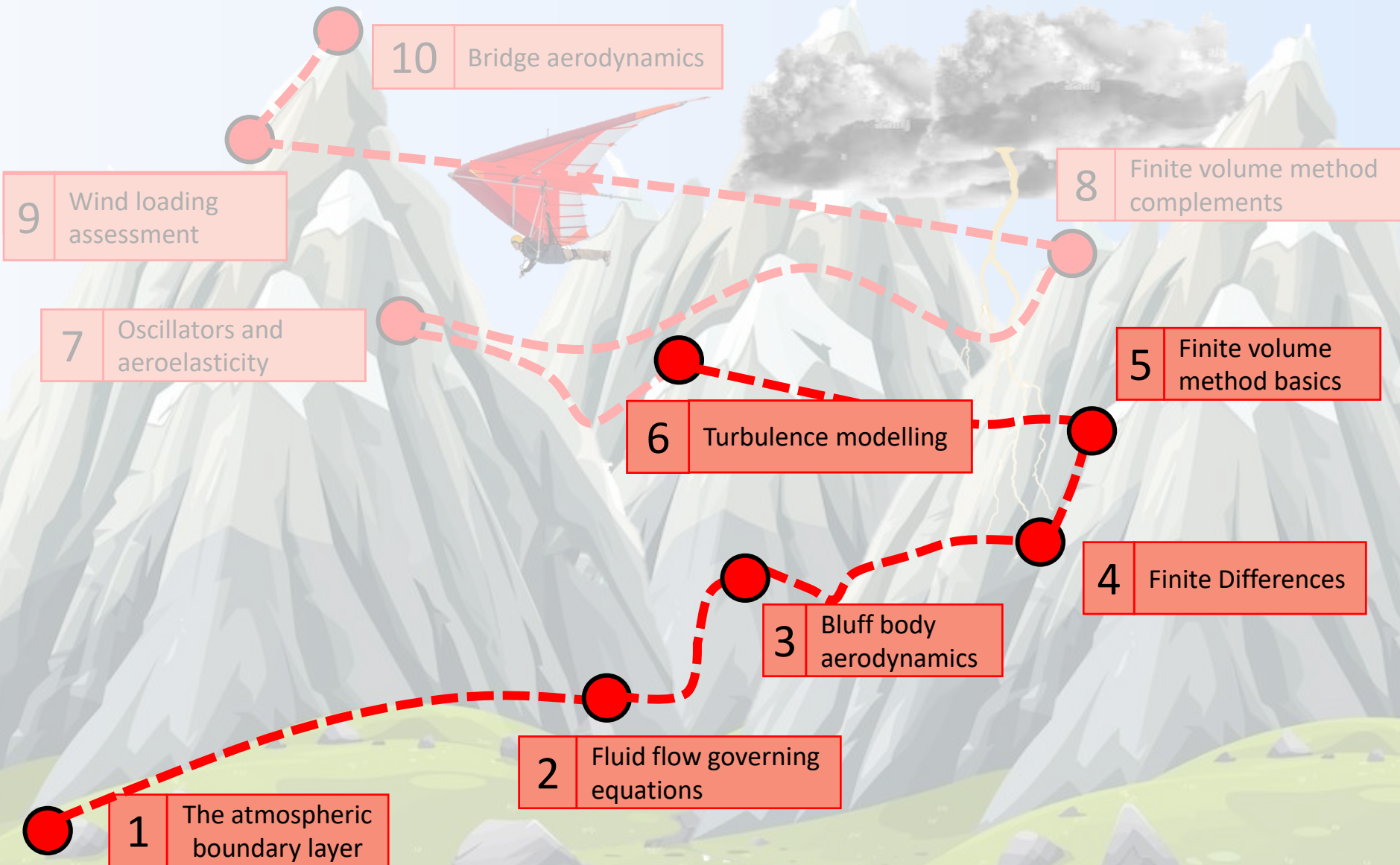
- The need for turbulence modelling
- Scale-resolving vs non scale-resolving
- RANS equations
- Standard $k - \varepsilon$ and $k - \omega$ model
- Smagorinsky

Map of the course

Wind Engineering

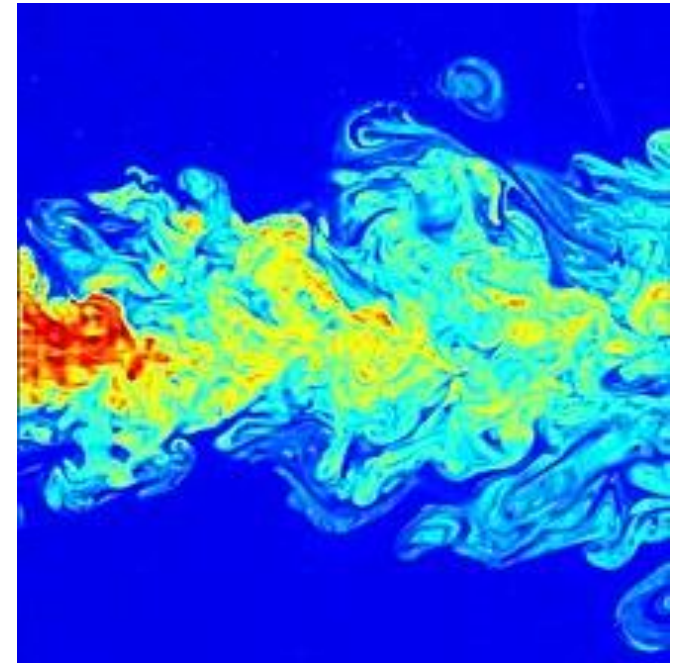
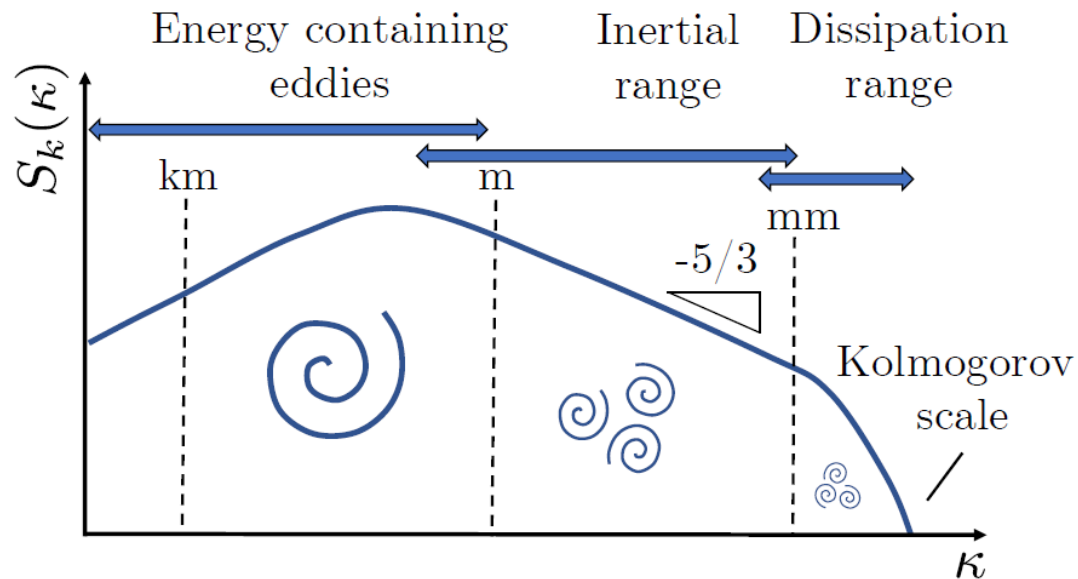
Fluid Dynamics

Numerical methods

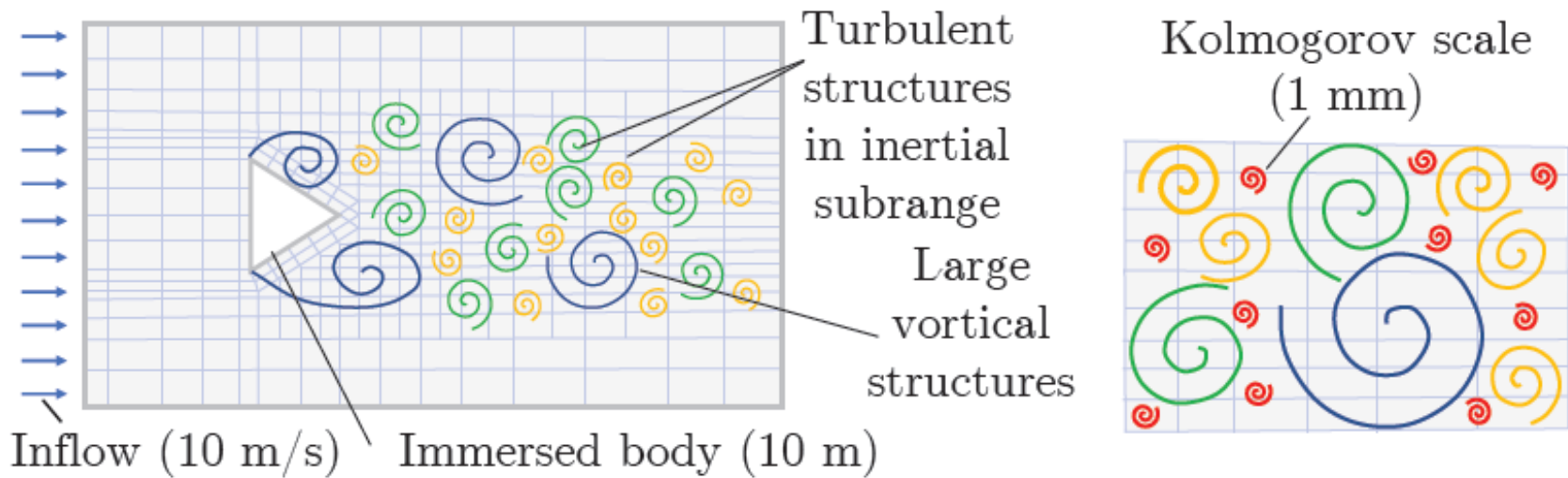


The need for turbulence modelling CWE

We already discussed regarding the multi-scale nature of turbulence. Simulating all scales requires huge (unachievable most of the times) computational efforts. **In addition, we are often not interested to such a level of detail.**



For instance, assume $L = 10 \text{ m}$, $I_u = 10\%$ and $\nu = 1.5E - 5 \text{ m}^2/\text{s}$.



After appropriate calculations, the Kolmogorov scale is found to be

$$l_K = (\nu^3 / \varepsilon)^{1/4} \quad \Rightarrow \quad 0.43 \text{ mm}$$

When salving up to the Kolmogorov scale, simulations are denoted as Direct Numerical Simulations (DNS).

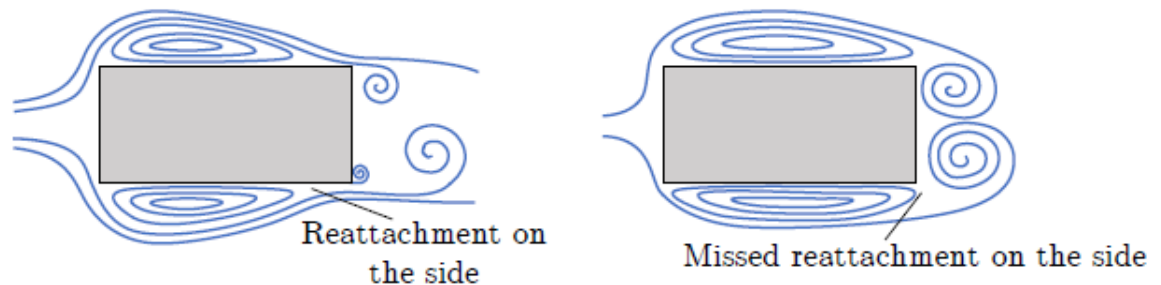
We however saw that turbulence, to some extent, can be regarded as an added viscosity. We now detail how turbulence models are built.

The simplest possible approach is....doing nothing!

$$\mu_e = \mu + \cancel{\mu_t} + \mu_{num} + \mu_{grid}.$$

...and hope that numerical and grid viscosity will lead to a reasonable representation of the effect of turbulence.

The approach is rather optimistic, and susceptible to spectacular failures, especially considering that the flow arrangement is often sensitive to details.

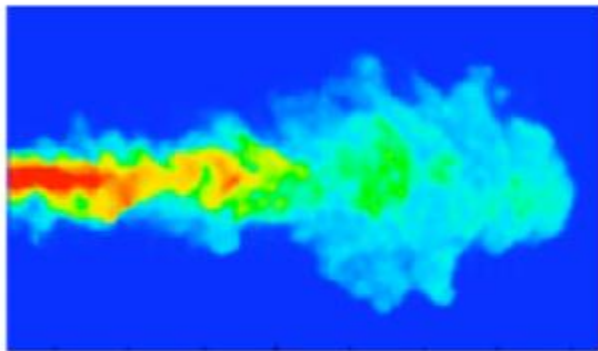


Note: the representation is extremely simplified and aimed at highlighting differences, not to be a representation of reality.

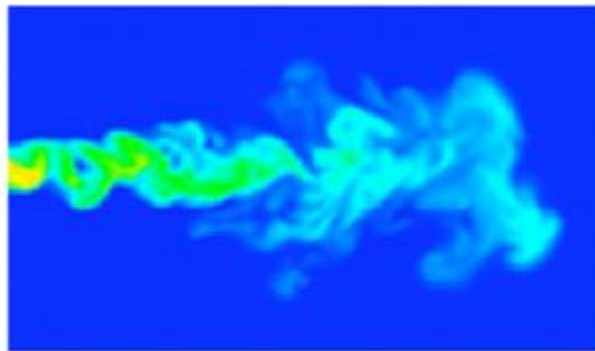
Instead of the optimistic approach of ILES, we might adopt a **Turbulence Model**. Turbulence models are mostly receipts used to calculate a local turbulent viscosity.

They do not necessarily represent reality faithfully and operate drastic surgery on the governing equations.

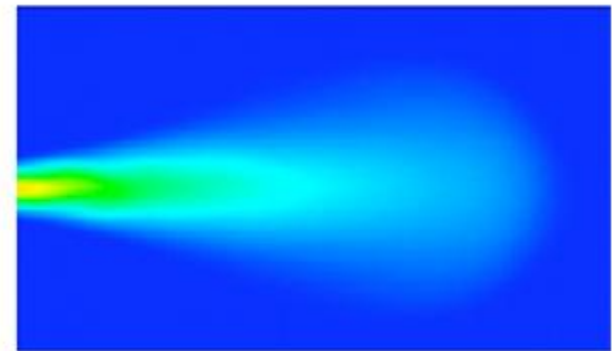
They greatly differ in expected accuracy, mode of utilization and computational costs.



Real/DNS



Scale-Resolving



Non Scale-Resolving

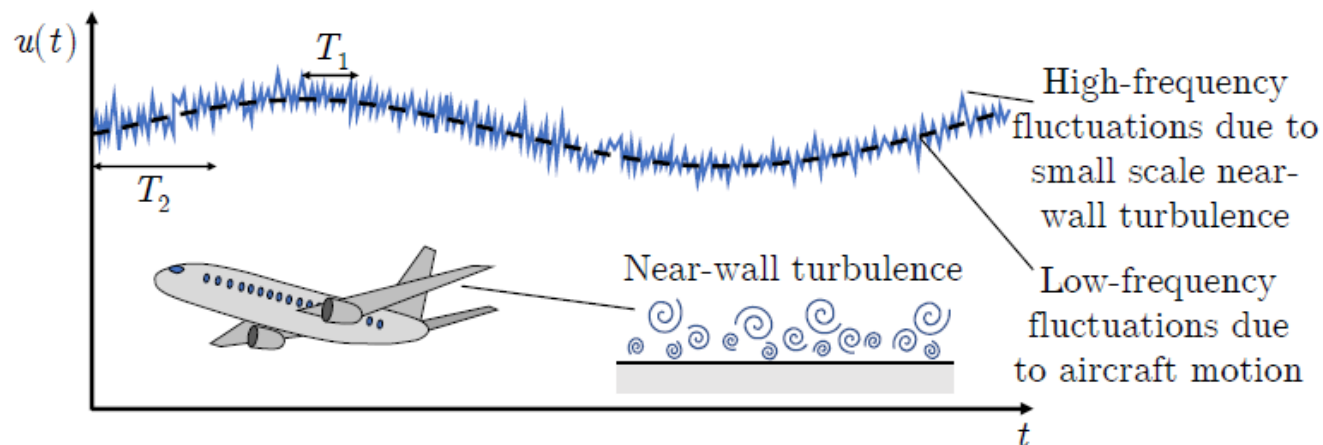
Image adapted from Som et al. Simulating flame lift-off characteristics of diesel and biodiesel fuels using detailed chemical-kinetic mechanisms and large eddy simulation turbulence model. Journal of Energy Resources Technology, courtesy of the authors, reproduced with the permission of ASME.

The first type of model we introduce is denoted as Reynolds-Averaged Navier-Stokes equations (URANS). **These models are usually non scale-resolving.**

For their development, we assume that a wide large scale separation exists between the overall flow and turbulence

T_1 is long compared to the fluctuations time-scale

T_2 is short compared to the mean flow evolution



The first step toward the development of the model is to **time-average the Navier-Stokes equations**. We adopt the usual splitting

$$u(x, t) = U(x, t) + u'(x, t),$$

with

$$U(x, t) = \frac{1}{T_1} \int_t^{t+T_1} u(x, t) dt, \quad u'(x, t) = u(x, t) - U(x, t),$$

The Navier-Stokes equation can be written in the form

$$\begin{cases} \nabla \cdot u = 0 \\ \frac{\partial u}{\partial t} + \nabla \cdot [uu^T] - \frac{1}{\rho} \nabla \cdot \sigma = \frac{f}{\rho}, \end{cases}$$

Let us start from mass conservation

$$\nabla \cdot [U + u'] = 0.$$

We time average over T_2 so that

$$\frac{1}{T_2} \int_t^{t+T_2} \nabla \cdot [U + u'] dt = 0.$$

Under appropriate regularity assumptions, the derivatives move out of the integral

$$\nabla \cdot \frac{1}{T_2} \int_t^{t+T_2} [U + \cancel{u'}] dt = 0,$$

$$\nabla \cdot \frac{1}{T_2} \int_t^{t+T_2} U dt = 0. \quad \Rightarrow \quad \nabla \cdot U = 0.$$

showing that the time-averaged field must be divergence-free.

We proceed analogously for the momentum equations

$$\frac{\partial[\mathbf{U} + \mathbf{u}']}{\partial t} + \nabla \cdot ([\mathbf{U} + \mathbf{u}'][\mathbf{U} + \mathbf{u}']^T) - \frac{1}{\rho} \nabla \cdot [\underline{\Sigma} + \underline{\sigma}'] = 0,$$

We time-average it

$$\frac{1}{T_2} \int_t^{t+T_2} \left[\frac{\partial[\mathbf{U} + \mathbf{u}']}{\partial t} + \nabla \cdot ([\mathbf{U} + \mathbf{u}'][\mathbf{U} + \mathbf{u}']^T) - \frac{1}{\rho} \nabla \cdot [\underline{\Sigma} + \underline{\sigma}'] \right] dt = 0.$$

The highlighted term can be expanded as

$$\frac{1}{T_2} \int_t^{t+T_2} \nabla \cdot [\mathbf{U}\mathbf{U}^T + \cancel{\mathbf{U}\mathbf{u}'^T} + \cancel{\mathbf{u}'\mathbf{U}^T} + \mathbf{u}'\mathbf{u}'^T] dt.$$

$$\frac{1}{T_2} \int_t^{t+T_2} \nabla \cdot ([\mathbf{U}\mathbf{U}^T + \mathbf{u}'\mathbf{u}'^T]) dt.$$

So that, starting from the original equation, the time-averaged momentum conservation equation reads

$$\frac{\partial[\mathbf{U} + \mathbf{u}']}{\partial t} + \nabla \cdot ([\mathbf{U} + \mathbf{u}'][\mathbf{U} + \mathbf{u}']^T) - \frac{1}{\rho} \nabla \cdot [\underline{\Sigma} + \underline{\sigma}'] = 0, \quad \text{(instantaneous)}$$



$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot ([\mathbf{U}\mathbf{U}^T]) + \nabla \cdot [\text{cov}(\mathbf{u}', \mathbf{u}')] - \frac{1}{\rho} \nabla \cdot \underline{\Sigma} = 0, \quad \text{(time-averaged)}$$

where we use the **covariance matrix between fluctuating velocity components**

$$\text{cov}(\mathbf{u}', \mathbf{u}') = \frac{1}{T_2} \int_t^{t+T_2} \begin{bmatrix} u'u' & u'v' & u'w' \\ v'u' & v'v' & v'w' \\ w'u' & w'v' & w'w' \end{bmatrix} dt,$$

which represents momentum fluxes due to turbulent motion. We denote these momentum fluxes as **Reynolds stresses**.

The time-averaged equations, i.e. the equations governing the time-averaged flow, are thus (disregarding source terms)

$$\begin{cases} \nabla \cdot \underline{U} = 0 \\ \frac{\partial \underline{U}}{\partial t} + \nabla \cdot (\underline{U} \underline{U}^T) - \frac{1}{\rho} \nabla \cdot [\underline{\Sigma} + \underline{\Sigma}_{Re}] = 0, \end{cases}$$

With great intuition and synthesis, using Boussinesq assumption

$$\underline{\Sigma} + \underline{\Sigma}_{Re} = -P \underline{I} + 2\rho \nu_e \nabla_s(\underline{U}),$$

where

$$\nu_e = \nu + \nu_t,$$

Turbulence models are mostly nothing else than reasonable receipts useful to calculate ν_t .

Actually, we already encountered a model able to provide a turbulent viscosity, i.e. Prandtl mixing length model which was used to derive the boundary layer equations.

In that case, mimicking molecular diffusion and heavily relying on dimensional analysis

$$l_t = \kappa_K y, \quad u_t = l_t \left| \frac{dU}{dy} \right|,$$

being y the distance from the wall.

Here we generally select

$$\nu_t = C_\mu l_t u_t.$$

being C_μ a constant to be calibrated and l_t and u_t some turbulent length and velocity scales. We need a recipe to calculate the latter.

A natural choice for the velocity scale related to turbulence is the **turbulent kinetic energy**

$$k = \frac{1}{T_1} \int_t^{t+T_1} \frac{(u')^2 + (v')^2 + (w')^2}{2} dt,$$

Starting from Navier-Stokes, after manipulation it, is possible to obtain

where

$$\frac{\partial k}{\partial t} + \nabla(k)U - \nabla \cdot (\nu_k \nabla(k)) = s_{k,p} - \varepsilon,$$
$$s_{k,p} = \frac{1}{\rho} \Sigma_{Re} : \nabla_s(U),$$
$$\varepsilon = s_{k,d} = \frac{1}{T_1 \rho} \int_t^{t+T_1} \underline{\sigma}' : \nabla_s(\underline{u}') dt = \frac{1}{T_1} \int_t^{t+T_1} 2\nu \nabla_s(\underline{u}') : \nabla_s(\underline{u}') dt,$$
$$j_{kt} = \frac{1}{T_1} \int_t^{t+T_1} \left(\frac{1}{2} |\underline{u}'|^2 + \frac{p'}{\rho} \right) \underline{u}' dt,$$

which cannot be easily evaluated, as they involve triple correlations of fluctuating quantities.

The problem is solved **assuming also in this case gradient diffusion**, adding σ_k as constant to be calibrated

$$\dot{j}_{kt} \approx -\frac{\nu_t}{\sigma_k} \nabla(k),$$

We thus reach

$$\frac{\partial k}{\partial t} + \nabla(k)U - \nabla \cdot (\nu_k \nabla(k)) = s_{k,p} - \varepsilon, \quad \nu_k = \nu + \nu_t / \sigma_k$$

notice that all terms in this equation can be evaluated relying on quantities measured on the time-averaged flow, except for ε , which represents a sink (destruction) of turbulent kinetic energy.

It is possible to obtain from Navier-Stokes an equation for ε which, however, involves even higher order correlations. **The problem is known as the closure problem.**

Based on the observation that the ingredient entering equations are always the same: **time variation, advection, diffusion, production and destruction**, we write

$$\frac{\partial \varepsilon}{\partial t} + \nabla(\varepsilon)U - \nabla \cdot (\nu_\varepsilon \nabla(\varepsilon)) = s_{\varepsilon,p} - s_{\varepsilon,d},$$

with

$$\nu_\varepsilon = \nu + \frac{\nu_t}{\sigma_\varepsilon}, \quad s_{\varepsilon,p} = C_{1\varepsilon} s_{kp} \frac{\varepsilon}{k}, \quad s_{\varepsilon,d} = C_{2\varepsilon} \frac{\varepsilon^2}{k},$$

Once ε is known, based on dimensional analysis, we can calculate

$$l_t = \frac{k^{3/2}}{\varepsilon}, \quad \longrightarrow \quad \nu_t = C_\mu l_t u_t = C_\mu \frac{k^{3/2}}{\varepsilon} k^{1/2} = C_\mu \frac{k^2}{\varepsilon},$$

It is thus possible to put all equations together

$$\begin{cases} \nabla \cdot \mathbf{U} = 0 \\ \frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot (\mathbf{U}\mathbf{U}^T) - \frac{1}{\rho} \nabla \cdot [\underline{\Sigma} + \underline{\Sigma}_{Re}] = \mathbf{0}, \end{cases}$$

where

$$\underline{\Sigma} + \underline{\Sigma}_{Re} = -P\mathbf{I} + 2\rho\nu_e \nabla_s(\mathbf{U}),$$

with $\nu_e = \nu + \nu_t$, being

$$\nu_t = C_\mu l_t u_t = C_\mu \frac{k^{3/2}}{\varepsilon} k^{1/2} = C_\mu \frac{k^2}{\varepsilon},$$

calculated based on

$$\frac{\partial k}{\partial t} + \nabla(k)\mathbf{U} - \nabla \cdot (\nu_k \nabla(k)) = s_{k,p} - \varepsilon,$$

$$\frac{\partial \varepsilon}{\partial t} + \nabla(\varepsilon)\mathbf{U} - \nabla \cdot (\nu_\varepsilon \nabla(\varepsilon)) = C_{1\varepsilon} s_{kp} \frac{\varepsilon}{k} - C_{2\varepsilon} \frac{\varepsilon^2}{k},$$

where we here avoid detailing any further the model for the sake of conciseness.

The $k - \varepsilon$ equation and its evolutions are known to behave well away from the wall, but poorly at the wall. Actually, it is possible to obtain interesting results with a model based on $\omega = \frac{\varepsilon}{k}$.

The equation for ω might be derived again from Navier-Stokes, presenting many unknown terms. More practically, we proceed as in the other cases

$$\frac{\partial \omega}{\partial t} + \nabla(\omega)U - \nabla \cdot (\nu_\omega \nabla(\omega)) = s_{\omega,p} - s_{\omega,d},$$

$$\nu_k = \nu + \alpha_k \nu_t, \quad \nu_\omega = \nu + \alpha_\omega \nu_t, \quad s_{\omega,p} = \gamma_\omega s_{k,p} \frac{\omega}{k}, \quad s_{\omega,d} = \beta_\omega \omega^2,$$

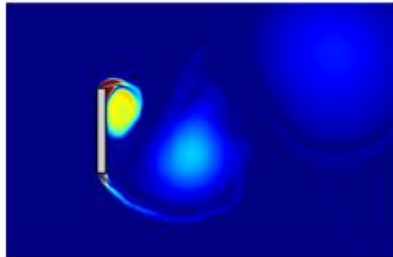
where production and destruction terms follow a logic identical to that used to develop the ε equation. It is thus possible to develop a $k - \omega$ model.

As it has been presented URANS attempt to model only the time-averaged velocity field. Numerous models exists which differ in their exact formulation and calibrated closure coefficients.

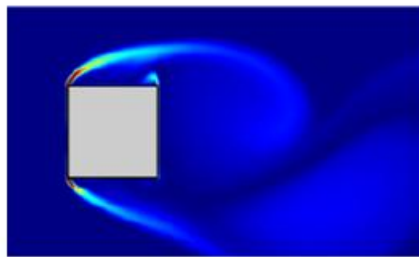
It is important to notice that the deduction of Reynolds-averaged equations is rigorous, but the operated simplifications are not.

The most commonly adopted models are:

- standard $k - \varepsilon$ model and its variations;
- standard $k - \omega$ model and its variations;
- $k - \omega$ SST a blending between the two (probably the most used model)



$R = 0.05$

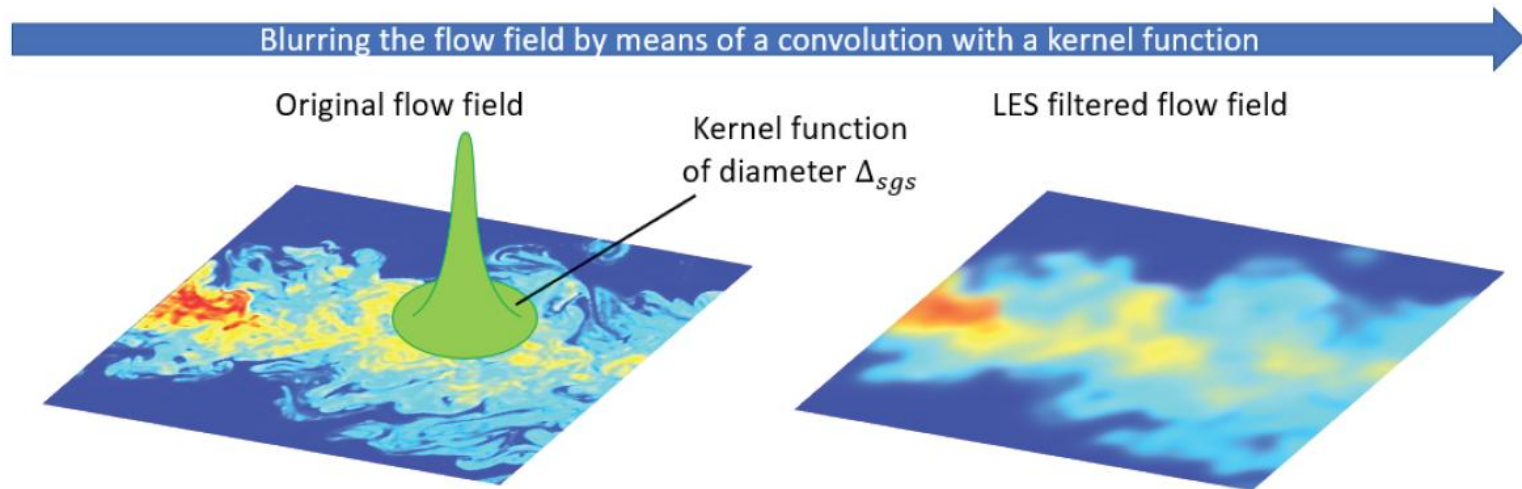


$R = 1.0$



$R = 5.0$

The most common models of the scale-resolving type are Large Eddy Simulations, LES. They are based on the space filtering of the Navier-Stokes equations



The decomposition now is based on

$$\mathbf{u}(\mathbf{x}, t) = \mathbf{U}(\mathbf{x}, t) + \mathbf{u}'(\mathbf{x}, t), \quad p(\mathbf{x}, t) = P(\mathbf{x}, t) + p'(\mathbf{x}, t),$$

being capital letters space-averages. After calculations, the same form of URANS is reached, where the stress term is modelled as

$$\underline{\underline{\Sigma}} + \underline{\underline{\Sigma}}_{sgs} = 2\rho(\nu + \nu_{sgs})\underline{\underline{\nabla}}_s(\mathbf{U}).$$

The equivalent of Reynolds stresses are now denoted as **subgrid stresses**. In fact, we adopt again

$$\nu_{sgs} \propto l_t u_t.$$

and set

$$l_t = C_s \Delta,$$

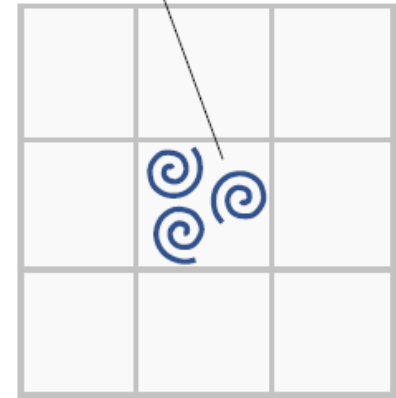
being Δ the local grid size, e.g. volume cubic root.

The model thus accounts for stresses (momentum fluxes) **induced by vortices which cannot be represented by the adopted grid**.

Then, similarly to the mixing length model we set

$$u_t = \left| \frac{dU}{dx} \right| l_t = \left| \frac{dU}{dx} \right| C_s \Delta.$$

Characteristic SGS
eddy size, $l_t = C_s \Delta$



The resulting turbulent viscosity is

$$\nu_{sgs} = C_s \Delta \left[C_s \Delta \left| \frac{dU}{dx} \right| \right] = (C_s \Delta)^2 \left| \frac{dU}{dx} \right|$$

In the general three-dimensional case, the equation above extends to

$$\nu_{sgs} = (C_s \Delta)^2 |\underline{\nabla}_s(\mathbf{U})|.$$

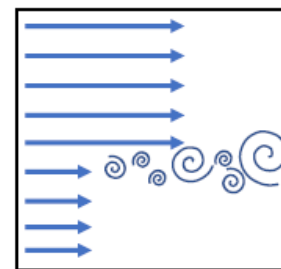
and C_s , the Smagorinsky constant, can be calibrated in standardized flows. In particular

Homogeneous-isotropic
turbulence decay



$$C_s \approx 0.17$$

Shear layers



$$C_s \approx 0.11$$

Also in the case of LES numerous variations are possible. Turbulence length scale is always related to the grid, while the velocity scale might be, for instance, derived by the k-equation (Yoshizawa one-equation model).

Essentially, LES allows to simulate part of the turbulent spectrum (large scales) and model the small scales.

